From Raw Data to Interactive Dashboard: A Data Analysis & Visualization Project

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ADAPTABILITY / ATTENTION TO DETAIL / COLLABORATION / DATA VISUALIZATION / DESIGN / ORAL PRESENTATION / PROBLEM SOLVING / PROJECT MANAGEMENT

Raw public health data is often too messy and complex for the general public to understand. Inspired by UBC's Making Research Accessible Initiative(MRAI), I worked with my team to transform the findings of a scholarly article on housing eviction and violence among people who inject drugs in Vancouver into an accessible, interactive data visualization. Below I walk through my entire process and the skills I demonstrated at each stage.

Background

Inspired by UBC's Making Research Accessible Initiative (MRAI), I worked with my team to transform the findings of a scholarly article by Kennedy, M. C., McNeil, R., Milloy, M.-J., Dong, H., Kerr, T., & Hayashi, K. (2017) on housing eviction and violence among people who inject drugs in Vancouver into an accessible, interactive data visualization.

Data Cleaning

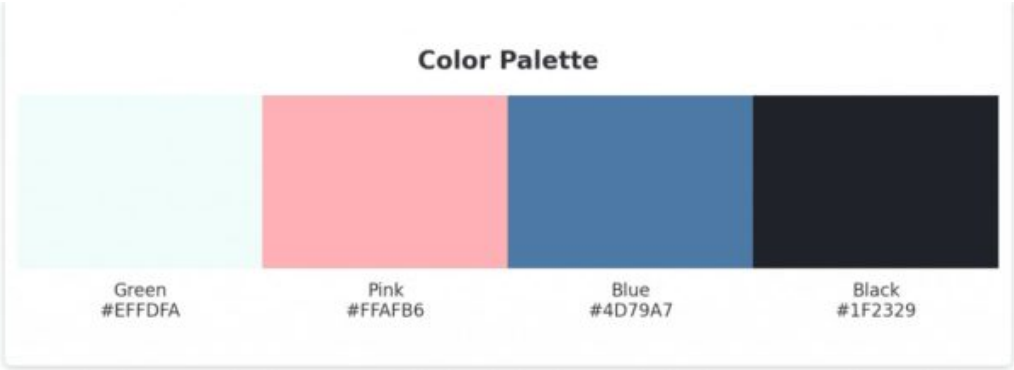
	Characteristic	Total n (%)	Males n (%)	Females n (%)	p-value	
2	Total	1689 (100)	1122 (66.4%)	567 (33.6%)	-	
3	Median age (IQR)	43 (37-49)	45 (38-50)	40 (33-46)	<0.001	
4	Caucasian ancestry	1000 (59.2)	734 (65.4)	266 (46.9)	<0.001	
5	‚à•High school educati	826 (48.9)	578 (51.5)	248 (43.7)		0.003
6	Childhood emotional al	693 (41.0)	426 (38.0)	267 (47.1)	<0.001	
7	In a stable relationship†	493 (29.2)	273 (24.3)	220 (38.8)	<0.001	
8	Downtown Eastside res	1103 (65.3)	713 (63.5)	390 (68.8)		0.033
9	Employed*	404 (23.9)	324 (28.9)	80 (14.1)	<0.001	
10	HIV seropositive	693 (41.0)	453 (40.4)	240 (42.3)		0.441
11	Residential eviction*	115 (6.8)	75 (6.7)	40 (7.1)		0.775
12	‚à•Daily crack cocaine	608 (36.0)	338 (30.1)	270 (47.6)	<0.001	
13	‚à•Daily heroin injectio	332 (19.7)	196 (17.5)	136 (24.0)		0.001
14	‚à•Daily cocaine inject	139 (8.2)	93 (8.3)	46 (8.1)		0.897
15	Heavy alcohol use*	281 (16.6)	179 (16.0)	102 (18.0)		0.275
16	Enrolment in addiction	949 (56.2)	603 (53.7)	346 (61.0)		0.005
17	Sex work involvement*	209 (12.4)	31 (2.8)	178 (31.4)	<0.001	
18	Drug dealing*	440 (26.1)	261 (23.3)	179 (31.6)	<0.001	
19	Incarceration*	222 (13.1)	170 (15.2)	52 (9.2)	<0.001	
20	Experienced violence*	356 (21.1)	245 (21.8)	111 (19.6)		0.282
21						
22	*Refers to 6 month period prior to interview					
23	‚Ä†IQR = Interquartile range					

First, I cleaned the raw data using Excel and Tableau Prep Builder. The original dataset provided by MRAI contained significant quality issues that made analysis impossible. I encountered and resolved numerous character encoding errors, including corrupted symbols like ",àû" and ",Ä†", as well as inconsistent percentage formats, unclear footnote placements, and misaligned variable labels. Using Excel and Tableau Prep Builder, I systematically cleaned the data by removing corrupted characters, standardizing numerical values and interquartile range notations, realigning table structures, and ensuring every data point corresponded to its correct column. As a result, the cleaned dataset became structurally consistent, analytically reliable, and ready for visualization. This step demonstrated my skills in data cleaning, data wrangling, attention to detail, and problem-solving with raw and messy data.

	Characteristic	Total	Males	Females
2	Total	1689	1122	567
3	Median age (IQR)	43	45	40
4	Caucasian ancestry	1000	734	266
5	High school education	826	578	248
6	Childhood emotional abuse	693	426	267
7	In a stable relationship*	493	273	220
8	Downtown Eastside residence*	1103	713	390
9	Employed	404	324	80
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11	Residential eviction*	115	75	40
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17	Sex work involvement*	209	31	178
18	Drug dealing*	440	261	179
19	Incarceration*	222	170	52
20	Experienced violence*	223	245	111

Colour Palette Design

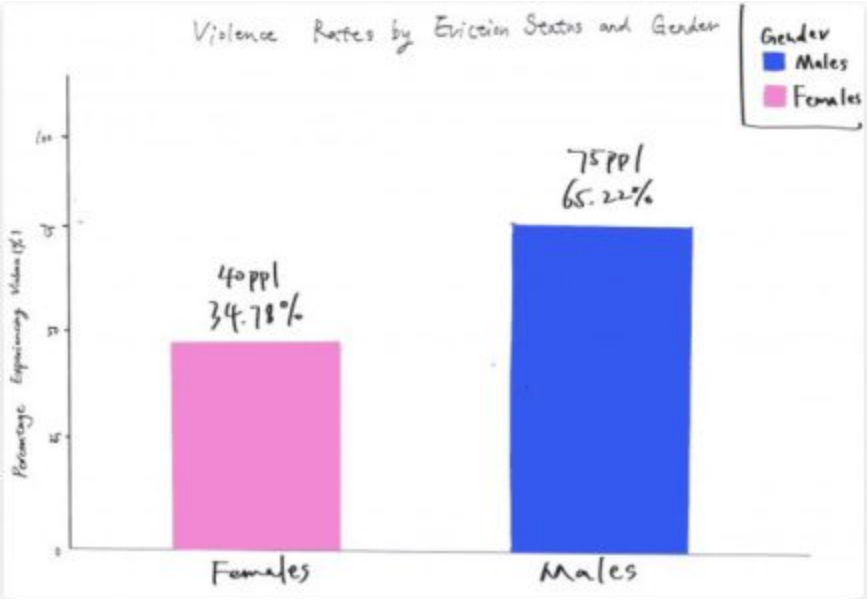
After cleaning the data, I established a colour palette to guide the visual design. I selected soft green (#EFFDFA), pink (#FFAFB6), deep blue (#4D79A7), and black (#1F2329) to balance clarity, emotional resonance, and accessibility. Effective visuals should evoke meaning and emotional connection. The pastel tones create a calm, inviting atmosphere, while the blue and black provide contrast and focus (Lankow, 2012). Guided by the "Do No Harm" principle, these colours also ensure legibility, minimize visual strain, and make the design inclusive and respectful of diverse viewers (Schwabish & Feng, 2021). This step improved my skills in colour theory, accessibility-aware design, visual communication.



Sketching & Chart Selections

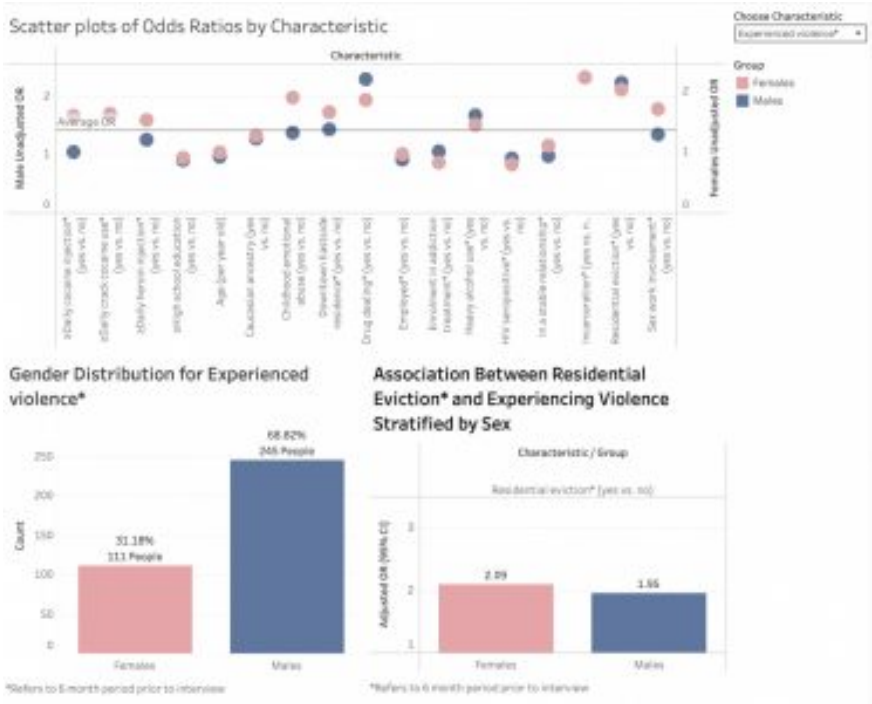
With the colour palette defined, I selected appropriate chart types based on the data and the story I wanted to tell. I created sketches to map out the visual hierarchy before moving into digital tools. For example, I sketched a bar chart to compare violence prevalence between genders. The sketch showed that among those who experienced violence, 65.22% were male and 34.71% were female; among those who did not experience violence, 75.00% were male and 40.00% were female. This early sketch helped me plan how to visually communicate gender differences in violence exposure. The key statistical measure I needed to visualize was the Adjusted Odds Ratio, which shows the relationship between residential eviction and violence while controlling for confounding factors such as sex work, police violence, and HIV status. The original study found that male PWID who experienced eviction were 1.95 times more likely to face violence, while

female PWID were 2.09 times more likely. I needed charts that would clearly illustrate this comparison for a general audience.



Dashboard Building

Using Tableau, my team and I built three core visualizations and integrated them into a single interactive dashboard. I created a bar chart presenting the adjusted odds ratios stratified by sex, with the Y-axis starting at one as the reference value, following best practices for odds ratio visualization. I created a scatter plot displaying unadjusted odds ratios across multiple predictors, using hue to distinguish male and female markers, which visually confirmed that residential eviction stands out as one of the most extreme predictors of violence for both genders. I also created a gender distribution bar chart showing violence prevalence, where 68.82% of those who reported violence were male, and 31.18% were female. The dashboard allows users to filter variables such as "Residential Eviction" or "Experienced Violence" and explore gender-based patterns dynamically. The use of filtering and highlighting interactions supports exploratory analysis and enhances user engagement (Munzner, 2014). This step demonstrated my skills in Tableau, interactive visualization, and user experience design.



Writing Report & Oral Presentation

After completing the visualizations, I led my team in writing a comprehensive report. The report covered our data cleaning methods, design choices, and theoretical foundations. These included Munzner's task abstraction framework, Tufte's data-ink maximization principle, and the "Do No Harm" framework. We also documented our iterative improvements based on peer feedback and reflected on the pros and cons of our work. I then delivered an oral presentation. I explained our process from raw data to the final interactive dashboard. I also answered questions from the audience.

Reference

Kennedy, M. C., McNeil, R., Milloy, M.-J., Dong, H., Kerr, T., & Hayashi, K. (2017). Residential eviction and exposure to violence among people who inject drugs in Vancouver, Canada. *International Journal of Drug Policy*, *41*, 59–64. https://doi.org/10.1016/j.drugpo.2016.12.017

Lankow, J., & Ritchie, J. (2012). Infographics: The power of visual storytelling. John Wiley & Sons. https://ebookcentral.proquest.com/lib/ubc/detail.action?docID=882721

Munzner, T. (2014). *Visualization analysis and design* (1st ed.). A K Peters/CRC Press. https://doi-org.ezproxy.library.ubc.ca/10.1201/b17511

Schwabish, J., & Feng, A. (2021). Do no harm guide: Applying equity awareness in data visualization. Urban Institute.

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LEARNING SIGNIFICANCE



Through this project, I learned to work effectively in a team. We divided tasks, shared feedback, and supported each other to complete the work on time. I also learned to use key tools. I used Excel and Tableau Prep Builder to clean messy data — fixing encoding errors, standardizing formats, and preparing the data for analysis. I used Tableau to build charts and create an interactive dashboard with filtering features.
Data analysis is not just about numbers — it is about asking the right questions and finding meaningful patterns. Data visualization is not just about making charts look good — it is about telling a clear, honest story that helps people understand complex information.

PROFILE

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Education

University of British Columbia

Faculty of Arts (2022 — present)

Media Studies Major

Work Experience

Unique Get Together Society (2024)

Marketing Team Member

Hebei Antu Trading Co., Ltd. *(2026)*

Social Media Operations Intern

Pinky Floy*(2023)*

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UBC Mood Psychology Club（2024 – Present）
Volunteer, Communications Member

SKILLS

Design (4)

Communications (3)

User-Centered Mindset (3)

Creativity (3)

Collaboration (3)

Attention to Detail (2)

Adaptability (2)

Time Management (2)

Building Connections (2)

Problem Solving (2)

Oral Presentation (2)

Storytelling (1)

Social Media Management (1)

Data Visualization (1)

Project Management (1)

COURSES

INFO250 (1)

INFO300 (1)

INFO419 (1)

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